Quadratics: Expanding & Factorising

Practice test 6 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify (x + 5)(x + 12)

[2 marks]

A2. Expand and simplify (x – 3)(x – 11)

[2 marks]

A3. Expand and simplify $(3x – 4)^2$

[2 marks]

A4. Fill in the two missing numbers: (x + 7)(x – ___) = x^2 + 2x – ___

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify (x – 3)(x – 2)(x + 6)

[3 marks]

B2. Expand and simplify $(x - 3)^3$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 3x - 54$

[2 marks]

C2. Factorise $x^2 - 17x + 72$

[2 marks]

C3. Factorise $16x^2 - 25$

[1 mark]

C4. Factorise fully $7x^2 - 7$

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $2x^2 - 9x + 10$

[2 marks]

D2. Factorise $4x^2 + 4x - 3$

[2 marks]

D3. Factorise $9x^2 - 12x + 4$

[2 marks]

Section E — Using factorising

E1. Solve $x^2 + 7x = 18$

[3 marks]

E2. Simplify fully

$$\frac{2x^2 + 5x - 3}{4x^2 - 1}$$

[3 marks]

Section F — Problem solving & proof

F1. A rectangle is $(2x + 3)$ cm long and $(x - 1)$ cm wide. Its area is 18 cm^2.
(a) Show that $2x^2 + x - 21 = 0$. (b) Solve the equation and find the perimeter of the rectangle.

[4 marks]

F2. Show that $(2x + 3)^2 - (2x - 3)^2 = 24x$ for every value of x

[2 marks]

F3. Without a calculator, work out $9.6^2 - 0.4^2$

[1 mark]